Embedded Systems Scripting

Engineering Technology

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

Embedded Systems Scripting (A11961)

Short Title:	Embedded Systems Scripting		
Faculty	Science and Computing		
Department	Engineering Technology		
Cluster	Software and Web Development		
Author	CLAIRE.FITZPATRICK		
Credits	5	**Level:**	Advanced

Description of Module / Aims

Embedded Systems Scripting module introduces the learner to the Linux operating system, commonly used command sets, bash shell scripting and a scripting language such as Python. The module will cover Python program syntax from the basic to the advanced to enable the learner design and develop complex scripts.

Programmes

	BEng (Hons) in Automation Engineering with Data Intelligence (`WD_EAUTO_B`)	3 / 5 / M
`EMBS-0002`	BEng (Hons) in Electrical and Automation Engineering (International) (`WD_ETRIC_BI`)	3 / 5 / M
`DASA-0001`	BEng (Hons) in Electronic Engineering (`WD_EONIC_B`)	3 / 5 / M
`EMBS-0002`	BEng (Hons) in Information Engineering (International) (`WD_EEELC_BI`)	3 / 5 / M
`EMBS-0002`	BSc (Hons) in Applied Computing (`WD_KACCM_B`)	3 / 5 / E
`EMBS-0002`	BSc (Hons) in Applied Computing (`WD_KCOMP_B`)	3 / 5 / E
`EMBS-0002`	BSc (Hons) in Computer Science (`WD_KCMSC_B`)	3 / 5 / E
`EMBS-0002`	BSc (Hons) in the Internet of Things (International) (`WD_KINTT_BI`)	3 / 5 / M

Indicative Content

- Introduction to Linux - Use of the command line, Customising the shell, File and directory management, Basic text editing, Useful commands and utilities, Text processing
- Introduction to Bash Shell Scripting - Basic bash script syntax, Variables, Conditionals, Loops, Shell commands
- Variables and Data Types
- Lists
- Basic Operators
- String Formatting
- Flow of Control
- Loops
- Functions
- Classes and Objects
- Dictionaries
- Modules and Packages
- Advanced Functionality - Generators, Decorators, Regular Expressions, Exception Handling, Sets, Serialisation, Partial Functions, Code Introspection, Closures

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Use the command shell to administer a Unix-like operating system, including basic shell commands, the Unix manual, and a text editor.
2. Write short shell scripts to automate simple repetitive and scheduled tasks.
3. Design complex scripts in a scripting language such as Python.
4. Develop complex scripts in a scripting language such as Python.

Learning and Teaching Methods

- This course will be presented by a combination of lectures and computer-based practicals whilst capitalising on a web-enhanced learning environment
- The lectures will be used to introduce new topics and their related concepts
- The emphasis on course delivery will be hands-on, problem solving both individually and in small class groups using problem based worksheets during the practical sessions
- Self-directed learning will be encouraged throughout the duration of the module

Assessment Methods

	Weighing		Outcomes Assessed
Continuous Assessment	100%		
Practical		80%	1,2,3,4
In-Class Assessment		20%	1,2,3,4

Assessment Criteria

70% - 100%: The learner has: attained the module learning outcomes at an excellent level; demonstrated a comprehensive knowledge of the associated subject matter; achieved an excellent level of the skills required for the subject matter; demonstrated the ability to carry out further investigation and problem solving associated with the subject matter.

60% - 69%: The learner has: attained the module learning outcomes at a very good level; demonstrated a detailed knowledge of the associated subject matter; achieved a very good level of the skills required for the subject matter.

50% - 59%: The learner has: attained the module learning outcomes at a good level; demonstrated a good knowledge of the associated subject matter; achieved a good level of the skills required for the subject matter.

40% - 49%: The learner has: attained the module learning outcomes at a basic level; demonstrated a basic knowledge of the associated subject matter; achieved a basic level of the skills required for the subject matter.

<40%: The learner has not: attained the module learning outcomes; demonstrated sufficient knowledge of the associated subject matter; demonstrated a sufficient level of the skills required for the subject matter.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	24	
Practical	24	
Independent Learning	87	

Essential Material

- Lutz, M. _Learning Python_. 5th. Sebastopol, CA: O'Reilly Media, 2013.
- McGrath, M. _Linux in easy steps_. 5th. England: In Easy Steps, 2010.

Supplementary Material

- McGrath, M. _Python in easy steps_. England: In Easy Steps, 2013.
- Shaw, Z. _Learn Python the hard way_. 3rd. Crawsville, Indiana: Addison Wesley, 2013.

Requested Resources

- Room Type: Computer Lab